

LAB EXPERIMENT RECORD

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
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SET 4: PRODUCT INVENTORY MANAGEMENT

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Aim

To visualize product inventory quantities using bar and stacked bar charts, analyze the relationship between product price and quantity available using a scatter plot, and develop an interactive Tableau dashboard.

PROGRAM 1: Bar Chart of Product Inventory

Question

Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

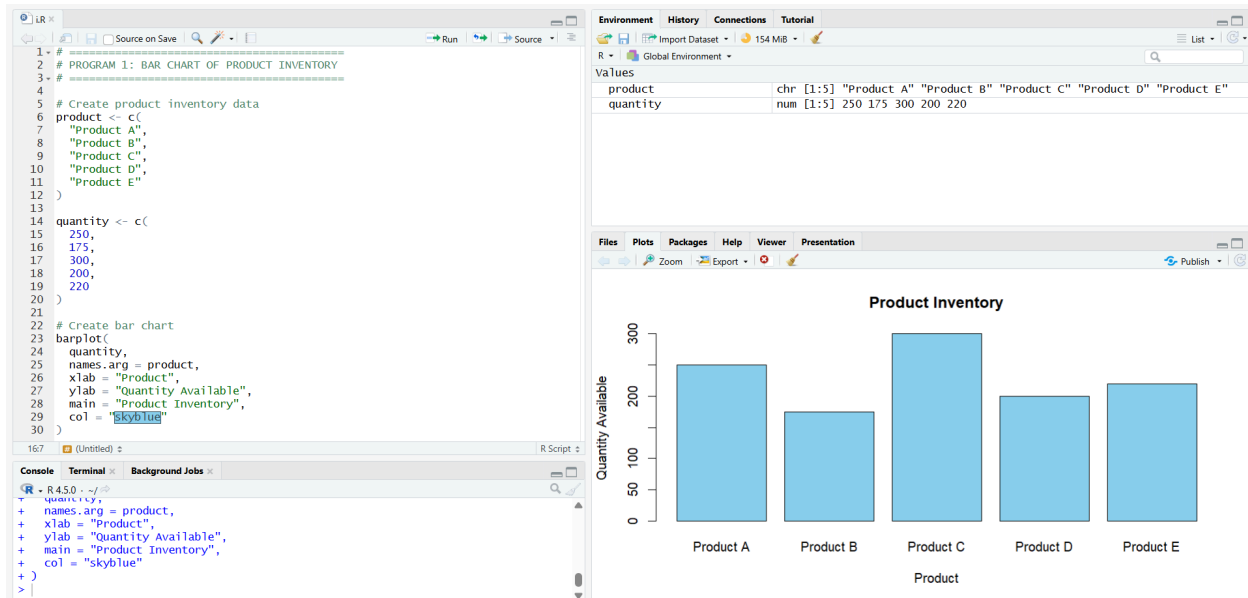
R Program

```
# =====  
# PROGRAM 1: BAR CHART OF PRODUCT INVENTORY  
# =====  
  
# Create product inventory data  
product <- c(  
  "Product A",  
  "Product B",  
  "Product C",  
  "Product D",  
  "Product E"  
)  
  
quantity <- c(  
  250,  
  175,  
  300,  
  200,  
  220  
)  
  
# Create bar chart  
barplot(  
  quantity,  
  names.arg = product,  
  xlab = "Product",  
  ylab = "Quantity Available",
```

```
main = "Product Inventory",  
col = "skyblue"
```

)

Output:



Observation

Product C has the highest inventory with 300 units, while Product B has the lowest inventory with 175 units.

PROGRAM 2: Stacked Bar Chart by Product Category

Question

Generate a stacked bar chart to show the quantity of each product within different product categories.

Important ⚠

Your original dataset contains:

- Product ID
- Product Name
- Quantity Available

It does not contain Product Category.

Therefore, to demonstrate this question, we'll use the following **sample category data**:

Product	Category	Quantity
Product A	Electronics	250
Product B	Electronics	175
Product C	Furniture	300
Product D	Furniture	200
Product E	Accessories	220

R Program

```
# =====  
# PROGRAM 2: STACKED BAR CHART  
# =====  
  
# Product data  
product <- c(  
  "Product A",  
  "Product B",  
  "Product C",  
  "Product D",
```

```
"Product E"  
)
```

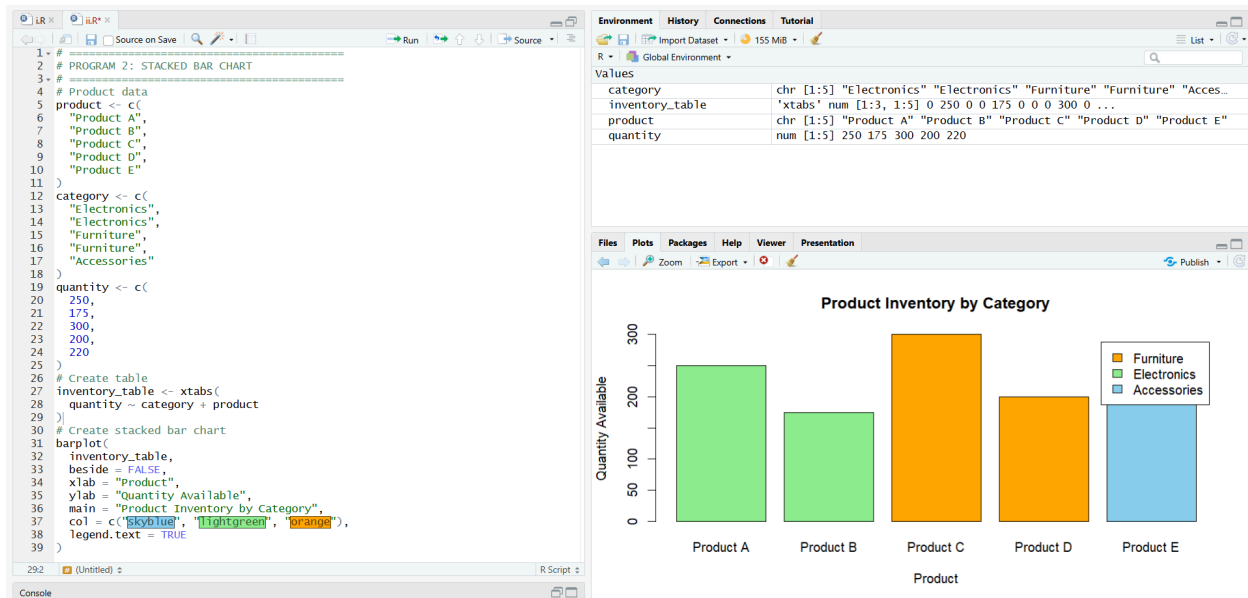
```
category <- c(  
  "Electronics",  
  "Electronics",  
  "Furniture",  
  "Furniture",  
  "Accessories"  
)
```

```
quantity <- c(  
  250,  
  175,  
  300,  
  200,  
  220  
)
```

```
# Create table  
inventory_table <- xtabs(  
  quantity ~ category + product  
)
```

```
# Create stacked bar chart  
barplot(  
  inventory_table,  
  beside = FALSE,  
  xlab = "Product",  
  ylab = "Quantity Available",  
  main = "Product Inventory by Category",  
  col = c("skyblue", "lightgreen", "orange"),  
  legend.text = TRUE  
)
```

Output:



Observation

The stacked bar chart shows how the available inventory is distributed among the different product categories.

PROGRAM 3: Scatter Plot — Product Price vs Quantity

Question

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

Important ⚠

Your original dataset does **not contain Product Price**.

Therefore, we'll use these **sample prices**:

Product	Price (\$)	Quantity Available
Product A	100	250
Product B	150	175
Product C	80	300
Product D	120	200
Product E	90	220

R Program

```
# =====  
# PROGRAM 3: SCATTER PLOT  
# =====  
  
# Sample product prices  
price <- c(  
  100,  
  150,  
  80,  
  120,  
  90  
)  
  
# Quantity available  
quantity <- c(  
  250,  
  175,  
  300,  
  200,  
  220  
)
```

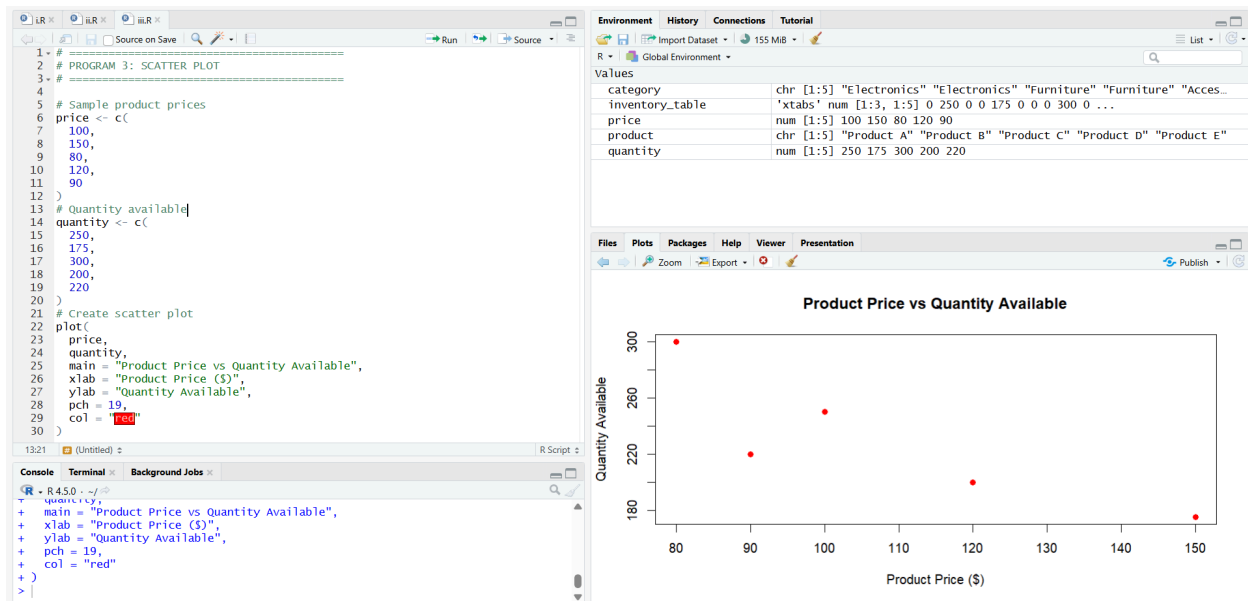
```

250,
175,
300,
200,
220
)

# Create scatter plot
plot(
  price,
  quantity,
  main = "Product Price vs Quantity Available",
  xlab = "Product Price ($)",
  ylab = "Quantity Available",
  pch = 19,
  col = "red"
)

```

Output:



Insight

The scatter plot shows a generally negative relationship between product price and quantity available in the sample data. Higher-priced products tend to have lower quantities available, while lower-priced products tend to have higher quantities.

PROGRAM 4: Tableau Dashboard

Question

Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

For Q4, we use **Tableau**, not RStudio.

Step 1: Prepare the Excel Dataset

For Tableau, create an Excel file containing:

Product ID	Product Name	Category	Quantity Available
1	Product A	Electronics	250
2	Product B	Electronics	175
3	Product C	Furniture	300
4	Product D	Furniture	200
5	Product E	Accessories	220

Save it as:

Product_Inventory.xlsx

The **Category** column uses the same sample category data from Q2.

Step 2: Open Tableau

Open **Tableau Desktop** or **Tableau Public**.

Select:

Connect → Microsoft Excel

Then select:

Product_Inventory.xlsx

Step 3: Create the Bar Chart

Create **Sheet 1**.

Drag:

Product Name



Columns

Then drag:

Quantity Available



Rows

Select:

Marks → Bar

Give the sheet the title:

Product Inventory Quantity

You should get approximately:

Product A 250

Product B 175

Product C 300

Product D 200

Product E 220

Step 4: Create the Stacked Bar Chart

Create a **new worksheet**.

Drag:

Category



Columns

Drag:

Quantity Available



Rows

Then drag:

Product Name



Color

The products will be stacked within their respective categories.

Give the worksheet the title:

Inventory by Product Category

Step 5: Create the Dashboard

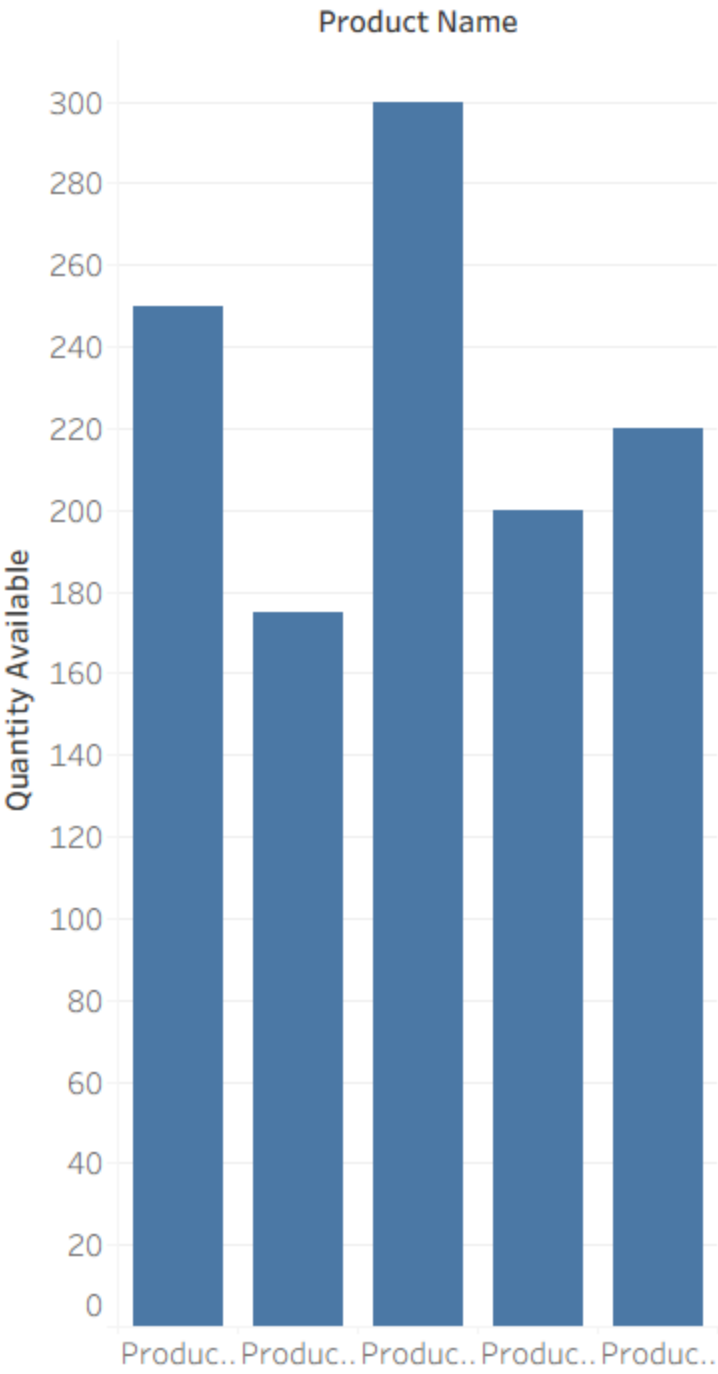
Click:

New Dashboard

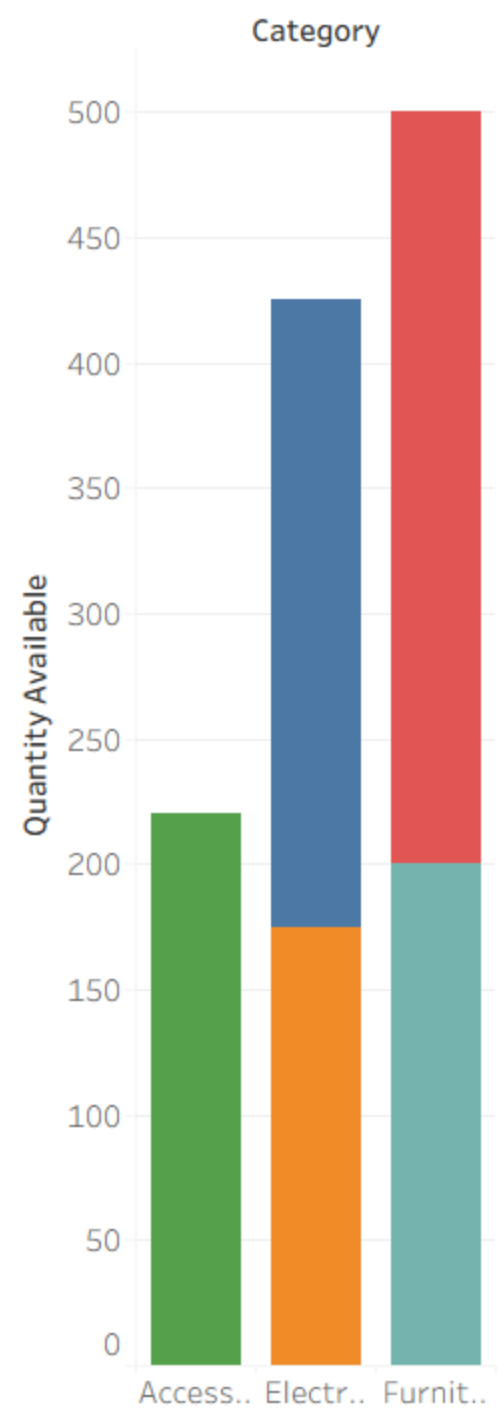
Add both worksheets.

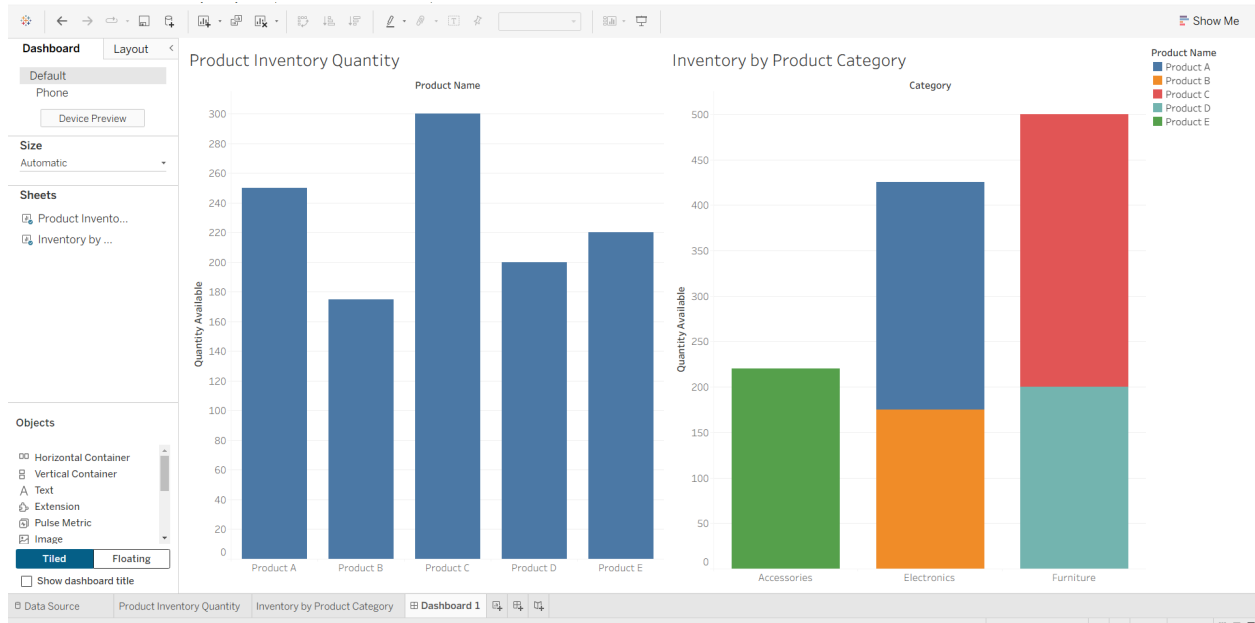
Your dashboard will look approximately like:

Product Inventory Quantity



Inventory by Product Category





Observation

The Tableau dashboard combines the product inventory bar chart and category-wise stacked bar chart. The interactive filters allow users to explore the inventory according to product categories.

Procedure

1. Create the product inventory dataset.
2. Store product names and quantities in R.
3. Use `barplot()` to create the product inventory bar chart.
4. Create sample product categories for the stacked bar chart.
5. Use `xtabs()` to create a category-wise product quantity table.
6. Use `barplot()` to create the stacked bar chart.
7. Create sample product price values for the scatter plot.
8. Use `plot()` to visualize the relationship between price and quantity.
9. Use `cor()` to calculate the correlation.
10. Prepare the inventory data in Excel.
11. Import the Excel file into Tableau.
12. Create the product inventory bar chart.
13. Create the category-wise stacked bar chart.
14. Create a new Tableau dashboard.
15. Add both charts to the dashboard.
16. Add filters/actions for interactive exploration.

Result

The product inventory data was successfully visualized using a bar chart and stacked bar chart. The relationship between product price and quantity available was analyzed using a scatter plot. An interactive Tableau dashboard was developed by combining the bar chart and stacked bar chart.